

What five modelling shortcuts cost a grid battery's revenue forecast



Four ordinary shortcuts inflate revenue by three to four times

Degradation pricing dominates; the efficiency curve is innocent

The five results

1	Pricing wear at zero	Implies 729 full cycles a year against the 346–451 a wear-aware schedule chooses	overstates 42–78%
2	No state-of-charge headroom for reserve	At £2/MW/h it commits 40.1 MW of reserve against 26.5 MW it could deliver	overstates 3–15%
3	Assuming you can forecast	A real forecast captures 60% of gross margin but only 48% of net — the gap widens as wear is priced higher	costs 40–52%
4	One flat round-trip efficiency	The two errors inside it have opposite signs and mostly cancel; what remains is the auxiliary load, not the curve	–2.9% to +5.7%
5	One wear price for every service	Peak-shifting ages a cell 1.8–1.9× faster than frequency response at equal throughput	shifts net 5.0–7.6%

Findings 1, 2, 4 and 5 compare two arms under perfect foresight — an unreachable bound used because both arms share it
Finding 3 is the one that measures what losing that foresight costs

One asset, one market, one window — so no finding gets its own flattering quarter

What is modelled, and what is not

Source: Elexon Insights (MID / APXMIDP), fetched at reproduce time

Modelled		Not modelled	
Asset	50 MW / 100 MWh lithium-iron-phosphate, 2 h duration 1 h and 4 h tested for finding 4	Day-ahead auction	The strategy is indexed to the half-hourly reference price; a real battery also trades the auction, where forecast error costs far less
Market	GB wholesale, half-hourly reference price plus a synthetic reserve-price sweep	Real reserve prices	The reserve sweep is synthetic, not market data
Window	1 March 2024 – 1 January 2026 672 execution days , one window for every finding	Network and fees	No use-of-system charges, no imbalance exposure
Degradation	Power-law calendar and cycle ageing, anchored to a field plant's measured loss over its own life	Other assets	One asset in one market: the shortcuts are shown to be costly here, and shown to be common by citation

Shorter windows in this dataset have wider spreads, so picking a quarter per finding would let each effect report itself at its best
Running everything on one window turned out to strengthen the results rather than soften them

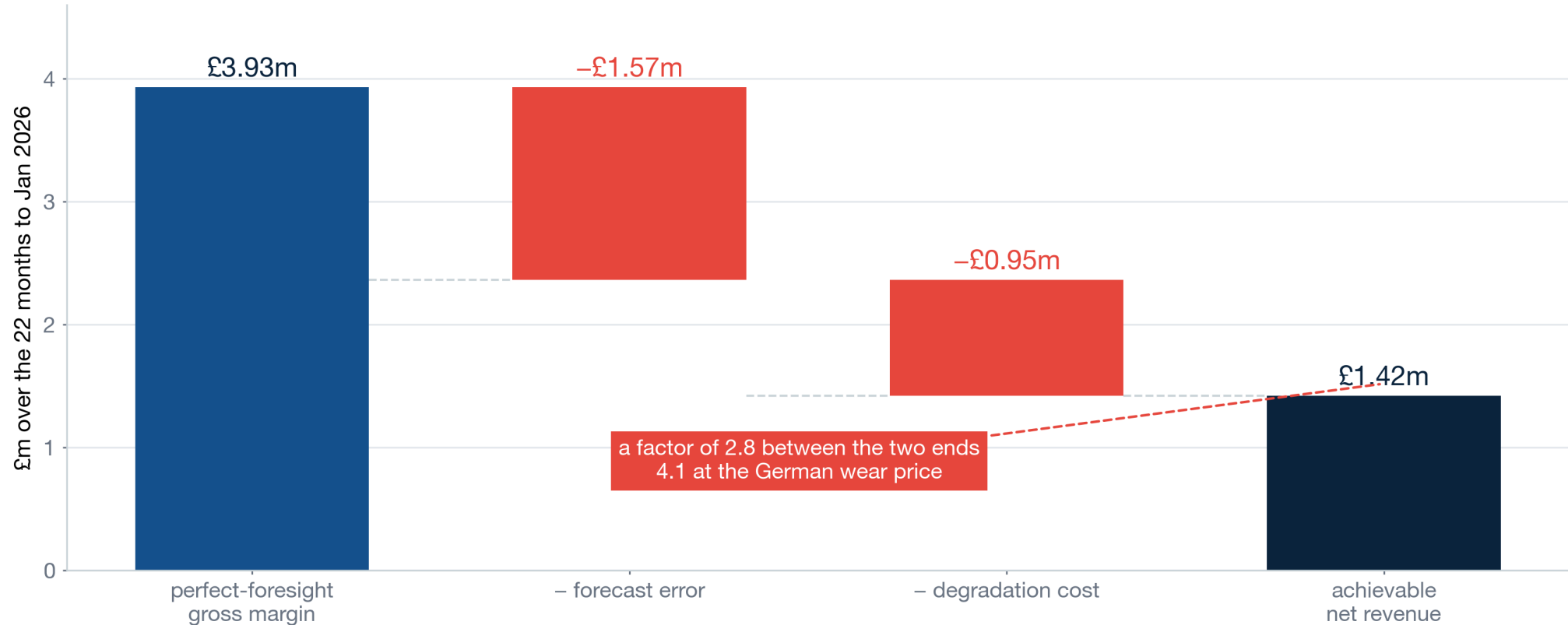
£3.93m of perfect-foresight margin leaves £1.42m

Forecast error costs more; wear pricing sets the ratio

From the theoretical ceiling to the achievable number, on the forecast-driven case

Unit: £m over 22 months

Source: Makuhari Development analysis



This is a wholesale-only strategy indexed to the half-hourly price, so £1.42m is the achievable revenue of the forecast-dependent leg, not of the asset

A model ignoring wear entirely would cycle 729 times a year and print £4.17m — the top bar already prices degradation

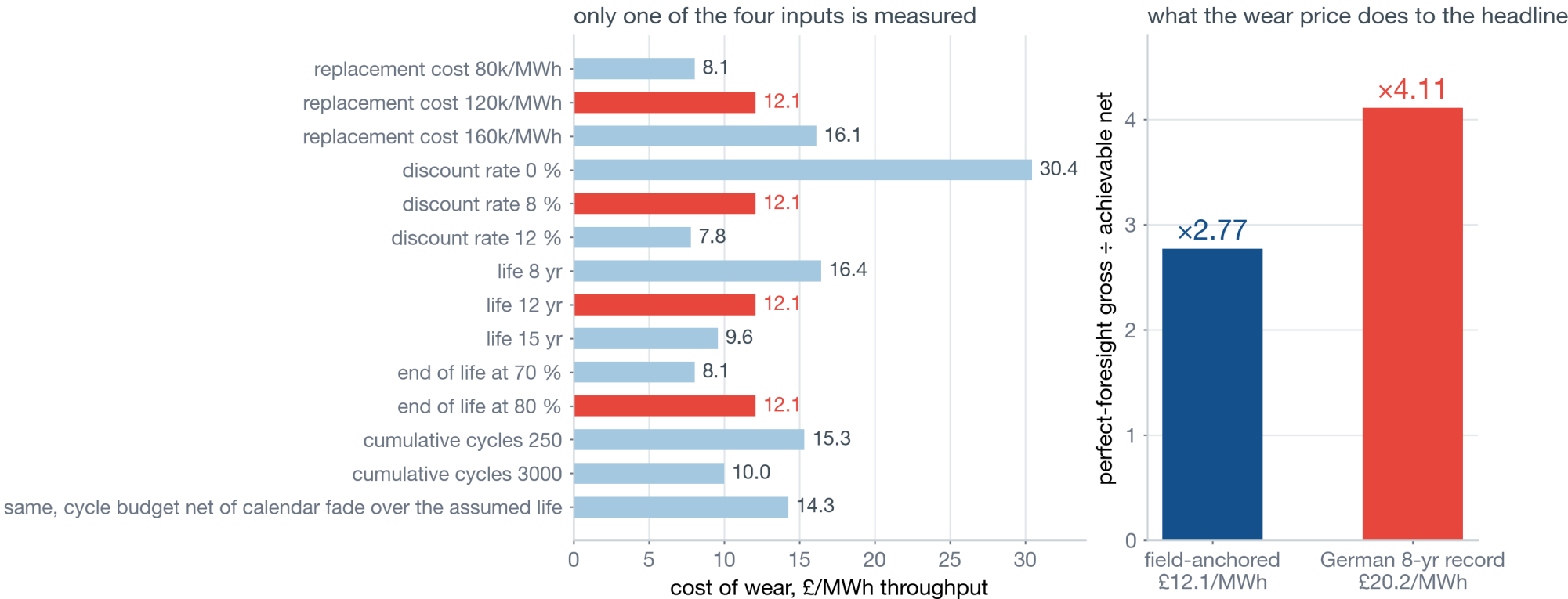
Shortcut 1: pricing wear at zero overstates net by 42–78%

The wear price has four inputs, and one of them is measured

Sensitivity of the cost of wear to its inputs, and what it does to the headline ratio

Unit: left £/MWh throughput, right ratio

Source: Makuhari Development analysis



Replacement cost, discount rate, assumed life and the end-of-life threshold each move the wear price by more than a third

Only the ageing model itself is anchored to measurement — to a field plant that lost 4.1% of capacity over 356 equivalent full cycles in three years

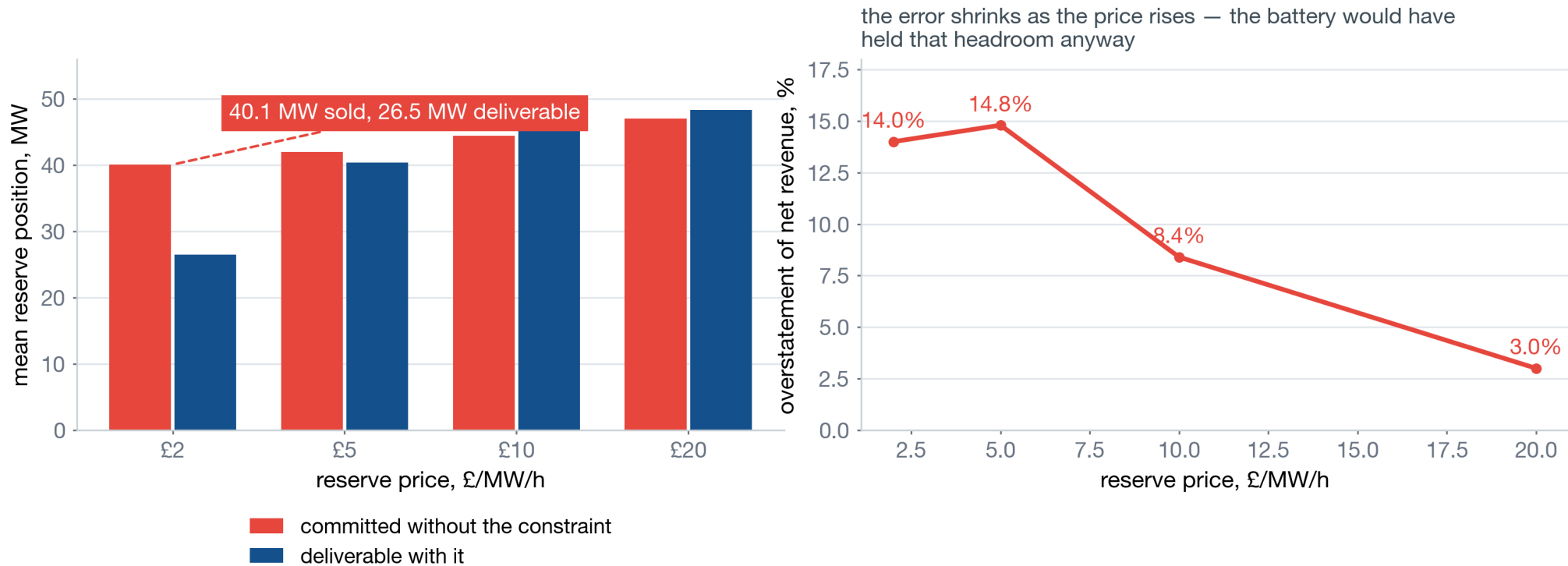
Shortcut 2: reserve sold without headroom cannot be delivered

The error is 3–15%, largest where the reserve price is lowest

Reserve position and the resulting overstatement across the price sweep

Unit: left MW, right %

Source: Makuhari Development analysis (reserve prices are a synthetic sweep, not market data)



At a high reserve price the battery holds energy back anyway, because reserve outbids arbitrage — so the constraint stops binding
The shortcut is therefore most dangerous in exactly the case a developer treats as conservative: a modest reserve price alongside an active arbitrage strategy

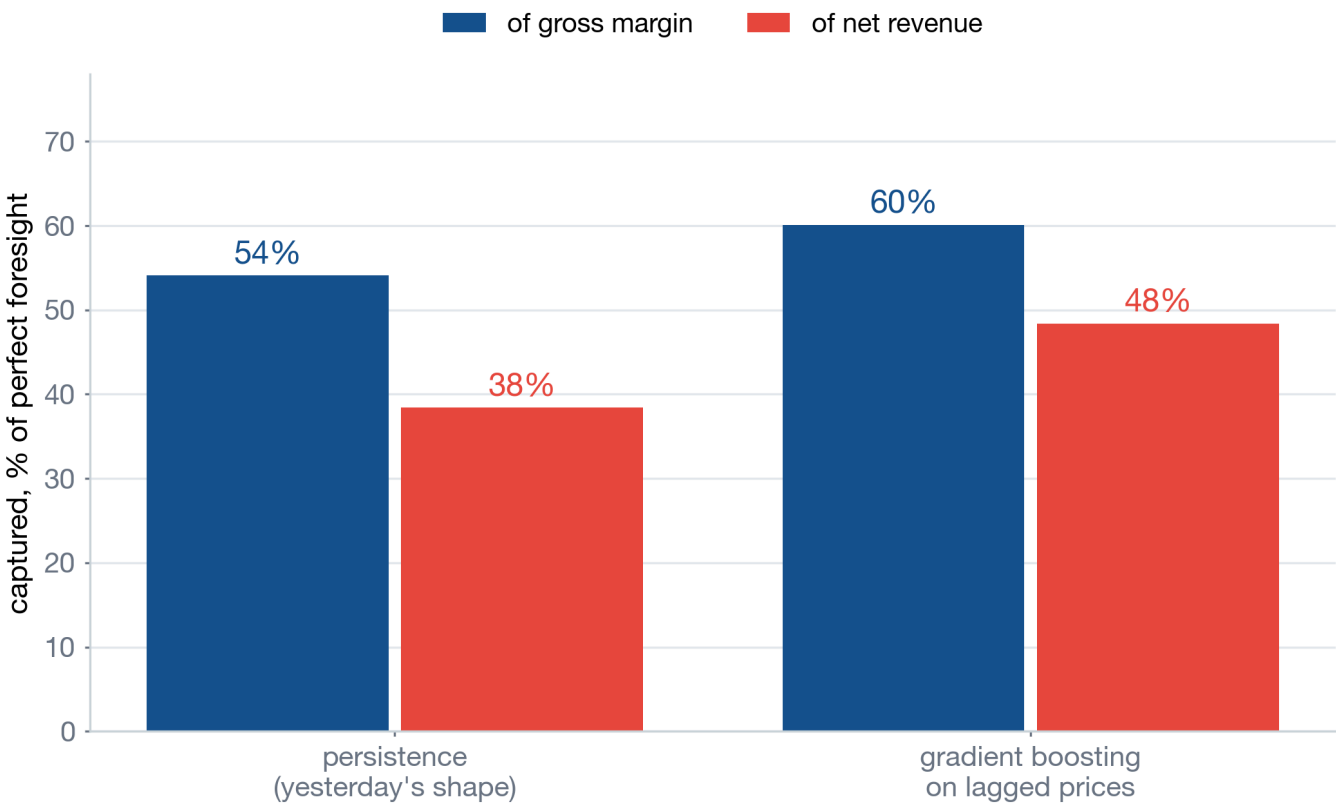
Shortcut 3: a forecast gets 60% of gross but 48% of net

The error lands on cycles you already paid wear for

Capture rate against perfect foresight, gross and net of the cost of wear

Unit: % of perfect foresight

Source: Makuhari Development analysis



- Gradient boosting on lagged prices beats persistence on both measures, but the **gross-to-net gap widens**: 1.24× at the field-anchored wear price, 1.54× at the German one
- The mechanism is not subtle. A mistimed cycle still consumes capacity, so the wear cost is paid whether or not the spread materialises
- This is a claim about **degradation-aware models**, not about forecasting: at zero wear price the two capture rates coincide
- Both predictors are deliberately naive — no weather, no outages, no fuel. A desk's forecast should beat them, which makes 48% a floor

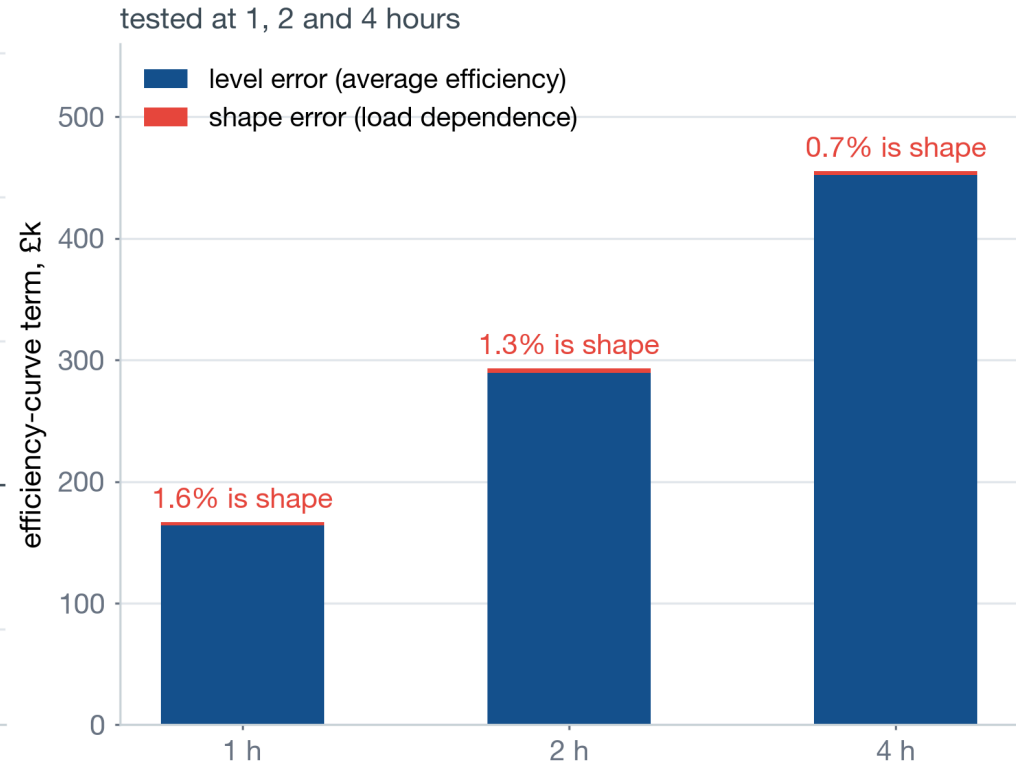
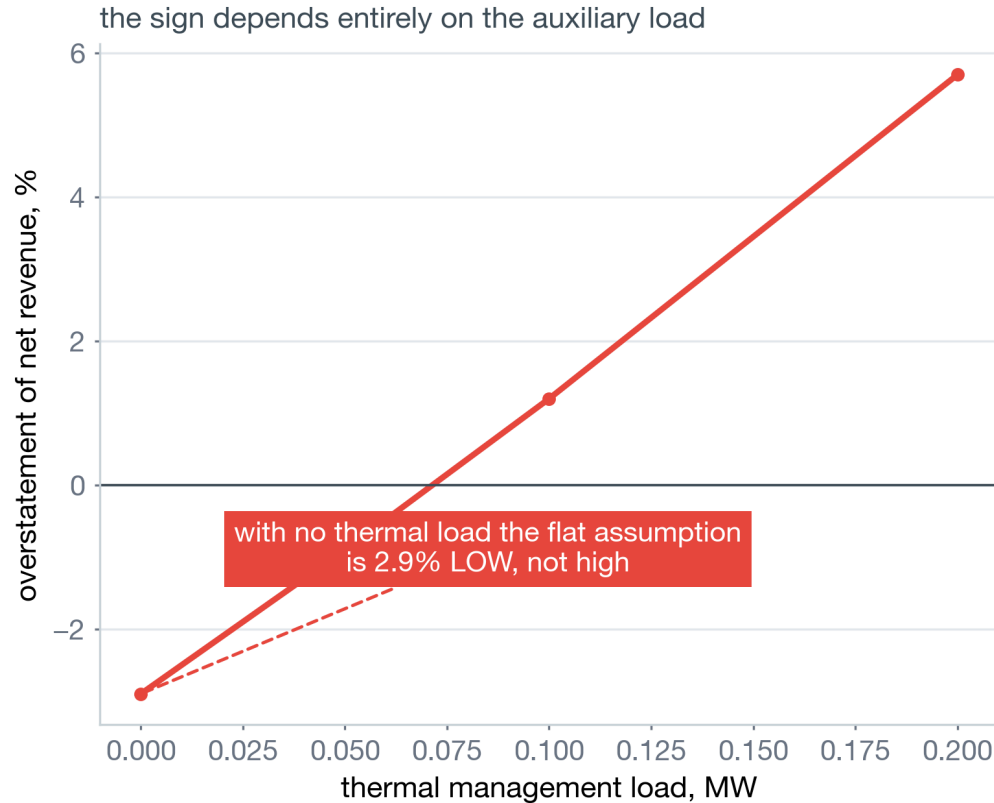
Shortcut 4: flat efficiency holds two errors of opposite sign

With no thermal load it reads 2.9% LOW, not high

Overstatement against thermal load, and the level / shape split of the efficiency term

Unit: left %, right £k

Source: Makuhari Development analysis



The efficiency half is a level error, not a shape one: equalising the average leaves load dependence worth 0.7–1.6% of the term, at 1, 2 and 4 hours alike

2 So the modelling effort belongs on where the auxiliaries are charged, not on resolving the converter curve — the opposite of where it usually goes

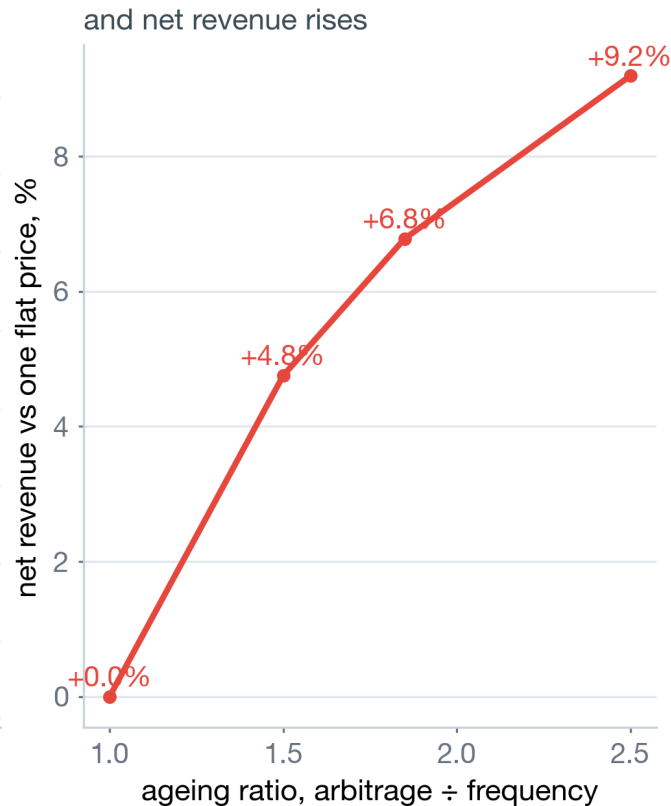
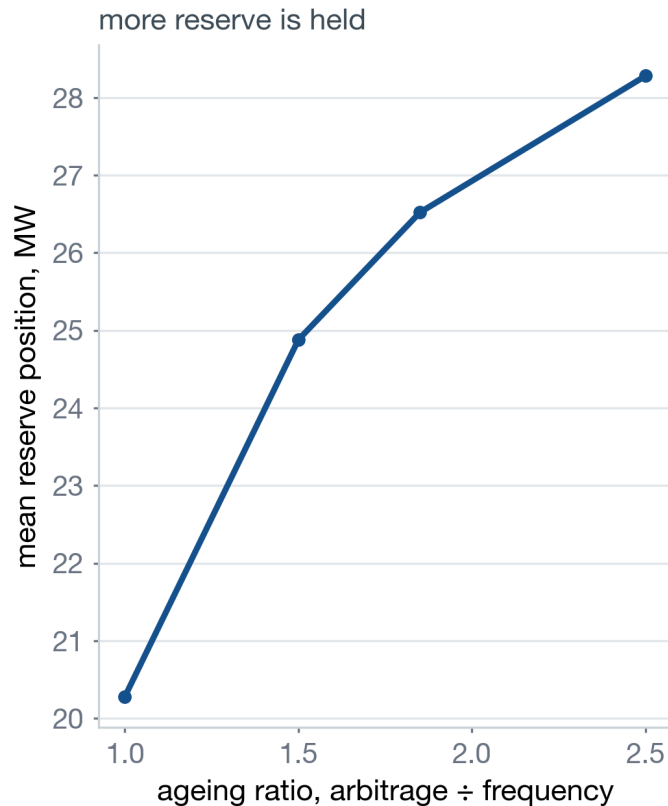
Shortcut 5: one wear price for every service costs 5–8% of net

Peak-shifting ages a cell 1.8–1.9× faster than reserve duty

Effect of differentiating the wear price by service, at a £2/MW/h reserve price

Unit: left MW, right % against a single flat price

Source: Makuhari Development analysis; ageing ratio from module tests on 220 Ah LFP under grid duty profiles



- Charging frequency response at the arbitrage wear price makes reserve look more expensive than it is, so the optimiser under-commits by **0.3–6.2 MW**
- Correcting it cuts cycling by 5.8–8.6% and raises net revenue by **5.0–7.6%** — the same energy, priced by what it actually costs
- It does **not** change whether the asset enters the reserve market at this wear price; at a higher one it would
- The assumption underneath — damage proportional to throughput — is exactly what degradation physics is known to violate, and is flagged in the code rather than buried

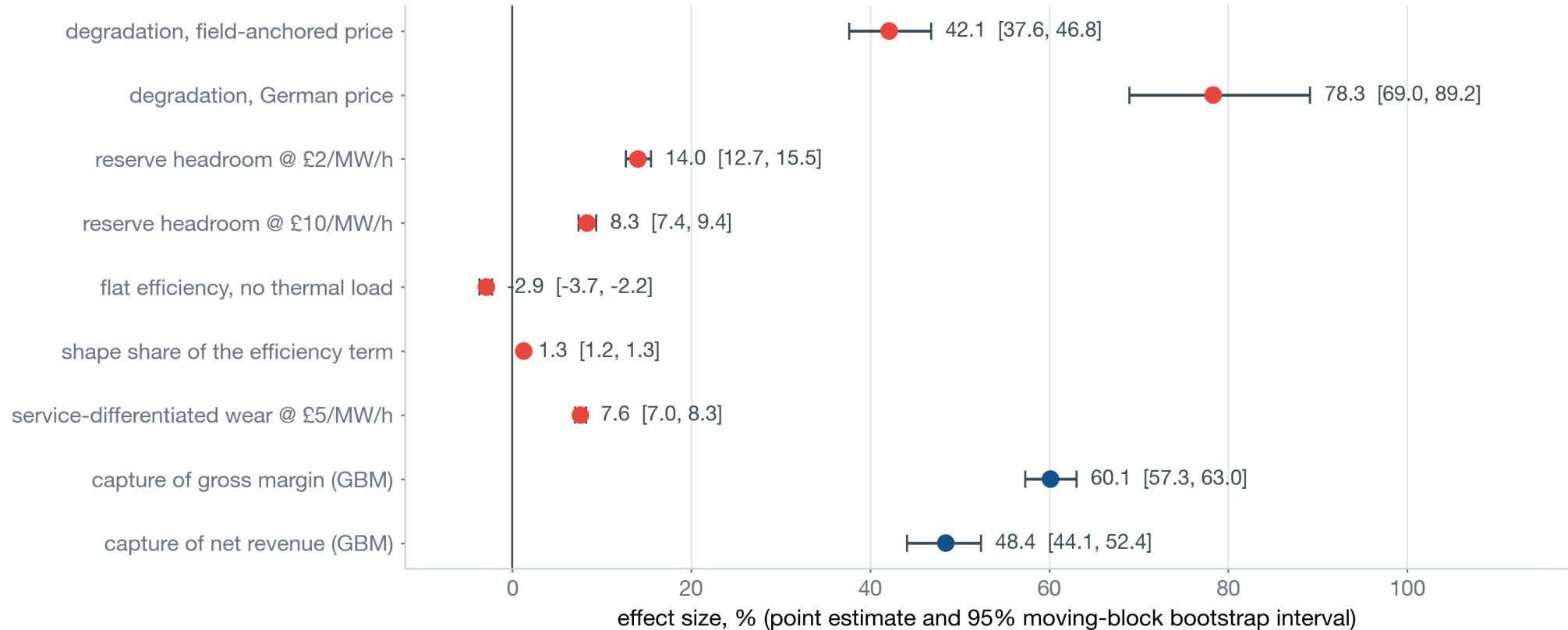
Every headline survives a moving-block bootstrap of the window

The degradation interval is the widest, and nowhere near zero

Point estimates with 95% intervals, resampling execution days in 7-day blocks (2,000 replicates)

Unit: %

Source: Makuhari Development analysis



Dispatch is solved once and the per-day results are resampled, so this measures sampling variation in the window — not uncertainty in the degradation model itself

Quarter-to-quarter variation is wider than the bootstrap interval in every case, which is the honest bound on transporting these numbers to another period

Three modelling choices behind these numbers, and their effect

Where this differs from the way battery revenue is usually modelled

Source: repository documentation and the cited field measurements

Choice	What it means	What it changes
Ageing as a power law, charged at the margin	Both ageing terms are power laws in their driver, with exponents below one, and only the cycle term enters the cost per MWh — calendar fade is a cost of ownership, not of trading	Read as linear rates, the same model predicts 67% capacity loss over the field plant's first three years against 4.1% measured. With the exponents applied, the field anchor is a 0.82 correction rather than a rescue
Auxiliaries separated from the efficiency curve	The converter's load-independent loss is charged only when it runs; thermal management is modelled as power × time and swept over 0–0.2 MW	Stops the same energy being charged twice, which is what makes the sign of finding 4 readable at all
No API keys, no manual downloads	The Elexon endpoints used need no key or registration; the client is self-contained and caches to parquet	One command reproduces every number from nothing. No market data is redistributed here

What to change in a revenue model, in order of what it is worth

The five findings as model changes

Source: findings 1–5

Change	Why	Worth	Effort
Price wear per MWh and put it in the objective	Without it the schedule cycles 729 times a year and the revenue line is a fiction	42–78% of net revenue	One term in the objective, four inputs to defend
Apply a capture factor to any perfect-foresight number	A real forecast reaches 48% of net, and the gap grows with the wear price	0.48–0.60 multiplier	None — it is a multiplier until you build the forecast
Constrain reserve by deliverable state of charge	Otherwise the model sells reserve the asset cannot deliver, hardest to notice at low reserve prices	3–15%	One coupling constraint per period
Differentiate the wear price by service	Peak-shifting ages the cell 1.8–1.9× faster than frequency response	5–8%	One extra coefficient
Model the auxiliary load — not the converter curve	The curve's shape term is worth about 1% of the efficiency term; the standing draw is the rest	up to 5.7% , sign depends on the auxiliaries	One binary per period

How to read this, and where the data comes from

How to read this

- This is a **controlled experiment on one asset in one market**, not a survey. The shortcuts are shown to be costly here, and shown to be common by citation
- It is a study of modelling error, **not investment advice**, and not a revenue forecast for any asset
- Reserve prices are a synthetic sweep. The wholesale strategy is indexed to the half-hourly reference price and excludes the day-ahead auction, where forecast error costs less
- Every number is reproducible from the public repository, including the intermediate results and the verification script

Sources

Source	What	Used for
Elxon Insights	GB half-hourly reference price (MID / APXMIDP)	All dispatch and revenue
Field plant, southern Italy	Measured capacity loss over three years	Anchoring the ageing model
German systems, 8-year record	Upper degradation rate	The high-wear scenario
Converter measurements, 11 setpoints	AC round-trip efficiency curve	Finding 4
Module tests, 220 Ah LFP	Ageing under grid duty profiles	Finding 5

No Elxon data is redistributed in the repository; each run fetches its own copy. Full citations for the field measurements are in the repository README.

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Data and code: github.com/ibarapascal/bess-revenue-stack

Companion report on the market side (JEPX, Japan): [jepx-revenue-anatomy](#)

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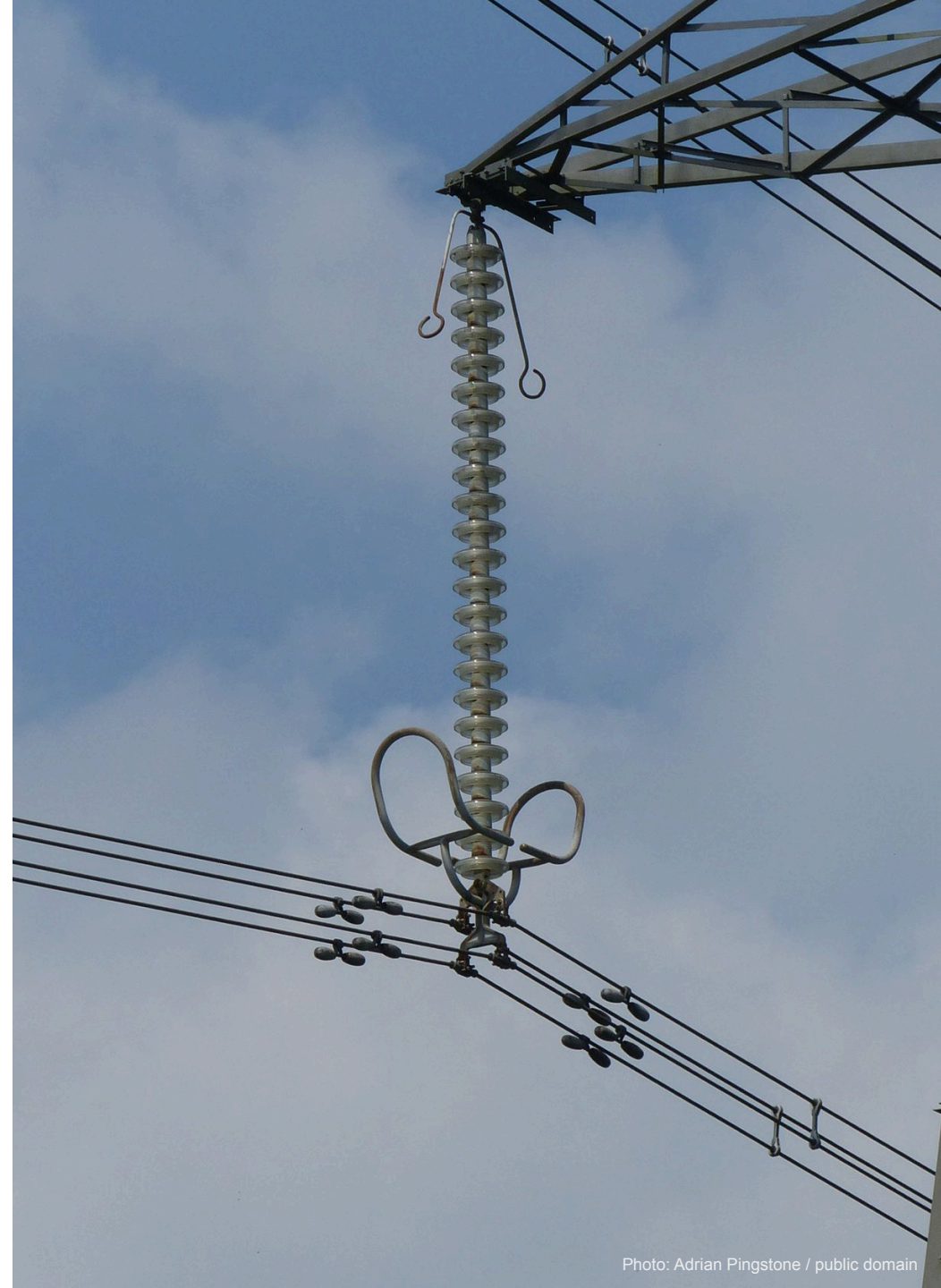


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